

# Unlabeled Bipartite Multigraphs: Asymptotic Expansions, Rare Symmetries and Connectivity

OEIS A007716 and its inverse growth (Part I);  
probability laws for A007716 and asymptotic expansions for A007718 (Part II)

Research notes prepared for Vladimir Reshetnikov with OpenAI

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## Abstract

**Part I.** Let  $a_n$  count nonnegative integer  $n \times n$  matrices of total sum  $n$ , modulo independent row and column permutations. Equivalently, these are bipartite multigraphs with  $n$  edges, no isolated vertices and distinguished shores. We prove a relative equivalent  $a_n \sim B_n^2 \exp(W(n)^2/2)/n!$ , where  $B_n$  is the Bell number, and a complete fixed-order expansion with rational functions of  $W(n)$  as coefficients. An exact invariant-partition formula controls all vertex automorphisms; the nonnegligible parallel-edge contribution is retained by a positive weighted count. We give the first two corrections, a finite coefficient algorithm, and all-orders smooth inverse models with rigorous integer-threshold brackets. The result strengthens the published logarithmic estimate for this sequence. No full classification of exponentially small sectors is claimed.

**Part II.** Choose uniformly an isomorphism class of bipartite multigraphs with  $n$  edges, distinguished shores and no isolated vertices. We prove that the probability of a nontrivial vertex automorphism is  $2W(n)^3/n + O((1+W(n))^7/n^2)$ . Outside probability  $O((1+W(n))^6/n^2)$ , every nontrivial vertex group is generated by one transposition of two leaves with a common neighbor. Disconnection has probability asymptotic to  $W(n)^2/n$  and is usually caused by one isolated edge. For the connected counting sequence A007718 we prove an all-orders finite-shift expansion in A007716, transfer the previously established complete relative asymptotic expansion, and give rigorous smooth inverse models and integer-threshold brackets. The proof distinguishes vertex automorphisms from parallel-edge permutations and controls graphs without a giant component. All orders are fixed, and no effective onset or priority claim is made.

**Editorial note (5 October 2026, batch 100).** This report is built from two manuscripts of ProveIt’s incoming-reports intake, Research Reports 88 and 92 of the session bundle that arrived in commit 60f54ea06 (manuscripts 01 and 02 of batch 100, placed in 36571ae0e). Report 88 (manuscript 01, 1 October 2026) is Part I, the base; Report 92 (manuscript 02, 2 October 2026) is Part II. Report 92 is a sequel, not a rival: it cites Report 88 as “the preceding report”, shipped its archive unchanged as a dependency, and uses its estimates as explicit inputs. Those inputs are printed once, in Part I; Part II’s Section 13 keeps manuscript 02’s restatement only where its form differs from Part I’s text, and refers to Part I otherwise. Part II answers the second of Part I’s further questions at leading order (the dated note in Section 10). Both manuscripts carry the author line “Research note prepared for Vladimir Reshetnikov with OpenAI”; the report is AI-assisted, unrefereed and not formalized. Part I is printed as delivered, and Part II with the symbol changes of Table 1, which also lists the symbols that each source reuses for two notions. Part I’s labels carry the prefix **bpm:**, Part II’s the prefix **bpm:sym:**. Notes added in this write

are marked “[Added 5 October 2026, batch 100: ...]”; Appendix A records the provenance and every editorial change.

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## Reading conventions and notation

Written 5 October 2026 (batch 100). Both Parts count the same objects:  $a_n$  is the number of isomorphism classes of bipartite multigraphs with  $n$  edges, no isolated vertices and two distinguished shores (OEIS A007716), with  $a_0 = 1$ ; isomorphisms preserve each shore, and transposition is not an extra identification. Throughout,  $w = W(n)$  is the principal (positive) Lambert function,  $we^w = n$ ,  $B_n$  is the Bell number,  $x = n/w = e^w$ ,  $\lambda = w^2/2$ , and  $\Phi$  is the explicit function of (27).

*Automorphisms.* In both Parts  $\text{Aut}(G)$  is the shore-preserving *vertex* group. Permutations of parallel edges are not automorphisms: they form the kernel of the map from the edge stabilizer in  $S_n$  onto  $\text{Aut}(G)$ , of order  $\Omega_G = \prod_{u,v} m_{uv}!$  (Part II, Lemma 14.1). The tempting reading “ $a_n \approx$  (number of edge-labelled realizations)/ $n!$ ” is false: the divergent factor  $e^{w^2/2}$  of Theorem 1.1 comes precisely from parallel edges.

*Connected counts.* Part II’s connected count is written  $c_n^\circ$  here (manuscript 02:  $c_n$ ), with the convention  $c_0^\circ = 0$ ; OEIS A007718 lists the conventional value 1 at index 0, and all positive indices agree. The bare letter  $c$  keeps Part I’s meaning:  $c(\sigma)$ , or  $c$  in  $B_c$  and  $d = n - c$ , is the number of cycles of  $\sigma \in S_n$ .

*Symbols.* Table 1 lists every symbol that the two manuscripts use for different notions, with the name printed here. Part I’s symbols are printed as delivered (its own double uses are declared in the table); every rename is in Part II, and Part II’s other symbols are manuscript 02’s. Where a display of Part II restates one of Part I, the reference points to Part I’s equation.

Table 1: Symbols with more than one meaning across the two manuscripts, and the names printed in this report. “02” is manuscript 02 as delivered.

| Symbol     | Part I (as delivered)                                                                     | Part II here (02’s symbol)                                                                                                                                                                           | Do not read as                                           |
|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| $R_n, R_1$ | $R_n(z) = \prod_{j < n} (1 + j/z)$ in (13); $R_1(w)$ the first correction of $Z_n$ , (25) | residual<br>$\mathcal{R}_n = a_n - Z_n - Y_{n-2}$ (02: $R_n$ )                                                                                                                                       | $\mathcal{R}_n$ is not Part I’s product $R_n(z)$         |
| $c$        | cycle count $c(\sigma)$ ; generic constants $c$ in tail bounds                            | $c(\sigma)$ as Part I; connected count $c_n^\circ$ (02: $c_n$ )                                                                                                                                      | $c_n^\circ$ is not a cycle count; $c_0^\circ = 0 \neq 1$ |
| $b$        | marked vector<br>$b = (b_2, b_3, \dots)$ ; $b(x)$ the logarithm of the Bell model         | reciprocal coefficients $\rho_j$ ,<br>$\sum \rho_j z^j = 1/\mathcal{A}(z)$ (02: $b_j$ )                                                                                                              | $\rho_j$ is unrelated to $b$ or $b(x)$                   |
| $D$        | cutoff $\lfloor A \log^2(n+1) \rfloor$                                                    | cutoff $D$ as Part I; ratio correction $\delta(w)$ in (55) (02: $D(w)$ )                                                                                                                             | $\delta(w)$ is not the cutoff                            |
| $A$        | cutoff constant $A, A_J$                                                                  | cutoff constant $A$ ;<br>generating function<br>$\mathcal{A}(z) = \sum a_n z^n$ (02: $A(z)$ );<br>$\alpha_1(w)$ (02: $A_1(w)$ )                                                                      | $\mathcal{A}$ is not the cutoff constant                 |
| $K, L$     | Dobinski variables $K, L$                                                                 | $K, L$ as Part I; kernel order $\Omega_G$ (02: $K_G$ ); fixed integer $q$ in Lemma 16.1 (02: $K$ ); constant $\tilde{C}_J$ in (60) (02: $K_J$ ); leaf-transposition count $\Lambda(G)$ (02: $L(G)$ ) | $\Lambda(G)$ counts transpositions, not leaves           |
| $E$        | weight $E(b)$ in (18); EGF $E(z)$ in Proposition 8.1; $\mathbb{E}$ expectation            | EGF $E^\circ(z)$ of $c_n^\circ$ (02: $E(z)$ )                                                                                                                                                        | $E(b)$ and $E(z)$ differ by argument                     |

| Symbol                          | Part I (as delivered)                                                                                       | Part II here (02's symbol)                                                                                                                                                        | Do not read as                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $\eta$                          | $\eta = C' \log n/n$                                                                                        | $\eta = Cw/n$ in (39),<br>attributed by 02 to Part I                                                                                                                              | the two forms are<br>equivalent since<br>$w \sim \log n$ ; Part I<br>proves the $\log n$ form<br>$x_J^\circ$ does not invert $F_J$ |
| $F_J, x_J, \varepsilon_J, N(y)$ | model $F_J$ for $a_n$ , its<br>inverse $x_J$ , width $\varepsilon_J$ ,<br>threshold $N(y)$                  | $F_J$ as Part I; connected<br>model $F_J^\circ$ (02: $G_J$ ), its<br>inverse $x_J^\circ$ (02: $x_J$ ), width<br>$\varepsilon_J^\circ$ , threshold $N^\circ(y)$ (02:<br>$N_c(y)$ ) |                                                                                                                                    |
| $C_J, C_j$                      | constant $C_J$ in (30);<br>constants $C, C'$                                                                | connected coefficients $P_j^\circ(w)$<br>(02: $C_j(w)$ ); exponent $M_J^\circ$<br>(02: $N_J$ )                                                                                    | $P_j^\circ$ are functions, not<br>constants                                                                                        |
| $M$                             | $M_{k,l}(n)$ ; moved support<br>$M$ ; exponents $M_J, M'_J$                                                 | 02's $M_J$ in its restated<br>expansion is Part I's $M'_J$ ;<br>unspecified exponents<br>$\xi_1, \xi_2, \xi_3$ (02: $M, M', M''$ )                                                | $\xi_i$ are existential, not<br>computed                                                                                           |
| $h$                             | number of selected cycles;<br>$h(k)$ the Stirling log-mass;<br>formal $h(X)$ for $a_n$                      | formal $h^\circ(X)$ for $c_n^\circ$ (02:<br>$h(X)$ )                                                                                                                              | $h^\circ$ uses $P_j^\circ$ , not $P_j$                                                                                             |
| $H, v$                          | $H = \sum_{i \geq 3} m_i$ ; long-cycle<br>defect $v$                                                        | fragment class $\Gamma$ (02: $H$ );<br>vertex count $ V(\Gamma) $ (02:<br>$v(H)$ ); entropy $\text{Ent}(p)$ (02:<br>$H(p)$ )                                                      | $ V(\Gamma) $ is not the defect<br>$v$                                                                                             |
| $r, N, F_n$                     | number of shores $r$<br>(Section 9); $N(y)$                                                                 | $\varrho(G)$ (02: $r(G)$ ); fixed onset<br>$n_0$ (02: $N$ );<br>$\Psi_n(t) = \Phi(t) + \Phi(n-t)$ (02:<br>$F_n(t)$ ); $g =  \text{Aut}(G) $ (02:<br>$m$ )                         | $\Psi_n$ is not the model $F_J$                                                                                                    |
| $T, \varepsilon$                | $T = KL$ ; Touchard $T_j$ ,<br>$T_k(\lambda)$ ; $\varepsilon$ in (14), in<br>Section 4, and $\varepsilon_J$ | $T = KL$ as Part I                                                                                                                                                                | the three $\varepsilon$ of Part I<br>are local                                                                                     |
| $u, a_i$                        | $a_i$ : Ewens exponents<br>in (20)                                                                          | 02's $u = W(x)$ in its<br>restated $\Phi$ is $w$ of (27); 02's<br>exponents $r_i$ are $a_i$                                                                                       | $a_i$ is not the sequence<br>$a_n$                                                                                                 |

*Scope of both Parts.* Neither source claims convergence of any infinite series, effective constants or numerical onset, a classification of exponentially small sectors or Stokes data, an unconditional single-ceiling inverse, or exhaustive priority; both state that the exact programs are regression evidence for finite identities and transcription, not proofs of the asymptotic tail bounds, and that neither is a Lean formalization or refereed. Part II adds that it did not re-audit Part I's analytic estimates (its **02-symmetry-verification.md**), that its reciprocal and Euler product are purely formal, and that its probabilities refer to the uniform measure on isomorphism classes. The statements each source leaves sketched or unproved are collected as questions in Sections 10 and 19.

## Part I

# Asymptotic expansions and inverse growth (Report 88)

## 1 The sequence and main results

Write  $a_n$  for the number of nonnegative integer  $n \times n$  matrices whose entries sum to  $n$ , up to separate row and column permutations. Deleting zero rows and columns identifies these objects with bipartite multigraphs with distinguished shores, no isolated vertices and  $n$  edges. Their first terms are

$$1, 1, 4, 10, 33, 91, 298, 910, 3017, 9945, 34207, \dots$$

This is OEIS A007716 [1]. Transposition is not an additional identification. Let  $B_n$  denote the Bell number and set  $w = W(n)$ , so that  $we^w = n$ .

**Theorem 1.1.** *As  $n \rightarrow \infty$ ,*

$$a_n \sim \frac{B_n^2}{n!} \exp(w^2/2) \sim \frac{\exp\{n(w - \log w - 1 + 2/w) + w^2/2 - 2\}}{(w+1)\sqrt{2\pi n}}. \quad (1)$$

The factor  $\exp(w^2/2)$  diverges. Keeping only the identity permutation in Burnside's lemma would miss it.

**Theorem 1.2.** *For every fixed integer  $J \geq 0$ , there are rational functions  $Q_j(w)$ , with  $Q_0 = 1$ , and a finite exponent  $M_J$  such that*

$$a_n = \frac{B_n^2}{n!} e^{w^2/2} \left\{ \sum_{j=0}^J \frac{Q_j(w)}{n^j} + O_J \left( \frac{(1+w)^{M_J}}{n^{J+1}} \right) \right\}. \quad (2)$$

*Section 5 gives a finite exact algorithm. The first correction is*

$$Q_1(w) = \frac{w^2(w^4 + 11w^3 + 22w^2 + 12w - 6)}{12(w+1)^2}. \quad (3)$$

*At order one, the remainder in braces is  $O((1+w)^{10}/n^2)$ .*

All orders are fixed as  $n$  tends to infinity. No convergence of the infinite series is asserted. In particular, this is stronger than a logarithmic equivalent.

[Added 5 October 2026, batch 100: the exponent  $M_J$  is shown to exist but not computed for  $J \geq 2$ , and the order-one bound  $(1+w)^{10}$  is asserted from three displayed error sizes (end of Section 6); both are recorded as Questions I.6 and I.7. Part II (Theorem 12.1 and Lemma 15.1) identifies the term  $w^3$  in  $Q_1 = R_1 + w^3$  as the leaf-transposition mass  $Y_{n-2}/Z_n$ . ]

Pavlichin, Jiao and Weissman [3, Theorem 12 and Appendix C] explicitly identify this sequence and prove  $\log a_n \sim n \log n$ . Their displayed result is  $\exp((1+o(1))n \log n)$ , despite the accompanying phrase “precise asymptotics.” The incidence-matrix results of Cameron, Prellberg and Stark [4, 5] concern zero-one matrices and different labeling conventions. We discuss this distinction further in Section 11.

## 2 Exact weighted counts and invariant partitions

Let  $M_{k,l}(n)$  be the number of nonnegative  $k \times l$  matrices of total  $n$  with no zero row or column. Define

$$Z_n = \sum_{k,l \geq 0} \frac{M_{k,l}(n)}{k!l!} = \sum_{[G]} \frac{1}{|\text{Aut}(G)|}, \quad (4)$$

where automorphisms preserve the shores. Orbit–stabilizer proves the second equality. Adding isolated vertices gives

$$\binom{kl+n-1}{n} = \sum_{i \leq k, j \leq l} \binom{k}{i} \binom{l}{j} M_{i,j}(n).$$

Summing nonnegative terms after division by  $k!l!$  yields

$$Z_n = e^{-2} \sum_{k,l \geq 0} \frac{\binom{kl+n-1}{n}}{k!l!}. \quad (5)$$

For  $n > 0$ , terms with  $kl = 0$  are zero.

[Added 5 October 2026, batch 100: the labelled analogue  $\sum_{k,l} M_{k,l}(n)$ , without the factor  $1/(k!l!)$ , is OEIS A120733 (nonnegative matrices of total sum  $n$  with no zero row or column); the collection report [generating-functions-and-asymptotics/oeis-sequence-asymptotics/a260700-parabolic-double-cosets](#) names it, with Kotěšovec’s asymptotic note on it, as a different sequence from its own count. No result transfers between the labelled and the unlabelled count. ]

A pair of set partitions of  $[n]$  represents a multigraph with edge labels  $[n]$ : label  $i$  joins its containing block in each partition. Removing edge labels is the simultaneous  $S_n$  action. Thus, if  $f(\sigma)$  counts the set partitions invariant under  $\sigma$ ,

$$a_n = \frac{1}{n!} \sum_{\sigma \in S_n} f(\sigma)^2. \quad (6)$$

**Lemma 2.1.** *If  $\sigma$  has  $c$  element cycles with lengths  $\ell_1, \dots, \ell_c$ , then*

$$f(\sigma) = \sum_{\pi \in \Pi_c} \prod_{S \in \pi} \sum_{k | \gcd(\ell_i : i \in S)} k^{|S|-1}. \quad (7)$$

Moreover, writing  $d = n - c$ ,

$$B_c \leq f(\sigma) \leq B_c \prod_i \ell_i \leq 2^d B_c, \quad Z_n = \frac{1}{n!} \sum_{\sigma \in S_n} B_{c(\sigma)}^2. \quad (8)$$

*Proof.* An orbit of  $k$  blocks consumes a nonempty group  $S$  of element cycles. Its length  $k$  divides each  $\ell_i$  in that group. Each cycle chooses a residue phase among the  $k$  blocks, and a common cyclic shift has no effect, giving  $k^{|S|-1}$  choices. The groups of consumed cycles partition  $[c]$ , proving (7). The  $k = 1$  terms give  $B_c$ . For any group, its divisor sum is at most  $\prod_{i \in S} \ell_i$ : if  $g$  is their gcd, the sum is at most  $g^{|S|}$ , including the case  $|S| = 1$ . Also  $\ell \leq 2^{\ell-1}$ . Finally, Dobinski’s identity  $B_c = e^{-1} \sum_{k \geq 0} k^c / k!$  and  $\sum_{\sigma \in S_n} x^{c(\sigma)} = (x)^{\overline{n}}$  turn the last sum in (8) into (5).  $\square$

## An elementary Bell ratio bound

For all  $j \geq 1$ ,

$$\frac{B_j}{B_{j-1}} \geq \frac{j}{16 \log(j+1)}. \quad (9)$$

Bell numbers are moments of a Poisson variable, so they are log-convex and  $B_j/B_{j-1} \geq B_j^{1/j}$ . For  $j \geq 2$ , put  $k = \lfloor j / \log(j+1) \rfloor$ . Then  $k \geq j / (2 \log(j+1))$ ,  $k \log k \leq j$ , and the corresponding term of Dobinski’s sum gives

$$B_j^{1/j} \geq k \exp\{-(1 + \log k!)/j\} \geq k e^{-1-1/j}.$$

This implies (9);  $j = 1$  is immediate.

Let  $s(n, n-d)$  be the unsigned Stirling number of the first kind. Since it equals  $e_d(1, \dots, n-1)$ ,

$$s(n, n-d) \leq \binom{n-1}{d} (n-1)_d \leq \frac{(n)_d^2}{d!}. \quad (10)$$

Every  $d$ -subset product is at most  $(n-1)_d$ , which proves the first inequality. Combining (8)–(10), the contribution of  $d > D$  in (6), relative to  $B_n^2/n!$ , is at most

$$\sum_{d>D} \frac{(4C^2 \log^2(n+1))^d}{d!}. \quad (11)$$

With  $D = \lfloor A \log^2(n+1) \rfloor$  and a sufficiently large fixed  $A$ , this is  $\exp(-c \log^2 n)$ .

### A quantitative comparison with the weighted count

Split off the nontrivial element cycles used by block orbits of length at least two. If  $h$  cycles are selected, their extra weight is at most  $B_h$  times their lengths' product. Their total length is at most  $2d$ . On  $d \leq D$ , equation (9) gives  $B_{c-h}/B_c \leq \eta^h$ , where  $\eta = C' \log n/n$ . Thus

$$1 \leq \frac{f(\sigma)}{B_c} \leq \sum_{h=0}^d \frac{B_h}{h!} (2d\eta)^h \leq \frac{1}{1-2d\eta} = 1 + O(\log^3 n/n).$$

Here  $B_h \leq h!$ , for example by writing each partition block as an increasing permutation cycle. Squaring, summing positive terms, and using (11) proves

$$\frac{a_n}{Z_n} = 1 + O(\log^3 n/n). \quad (12)$$

This bounds the weighted effect of automorphisms, rather than relying only on high-probability asymmetry.

## 3 The relative weighted saddle

Put  $x = n/w = e^w$ . Let  $K, L$  be independent with Dobinski law

$$\mathbb{P}(K = k) = \frac{k^n}{e B_n k!}, \quad k \geq 1.$$

Equation (5) gives exactly

$$Z_n = \frac{B_n^2}{n!} \mathbb{E} R_n(KL), \quad R_n(z) = \prod_{j=0}^{n-1} (1 + j/z). \quad (13)$$

For  $\varepsilon \leq 1/4$  with  $n\varepsilon^2 \gg \log n$ ,

$$\mathbb{P}(|K/x - 1| > \varepsilon) \leq \exp(-cn\varepsilon^2), \quad \mathbb{P}(K < x/2) \leq \exp(-cn). \quad (14)$$

Indeed Stirling reduces the logarithmic mass to  $h(k) - \frac{1}{2} \log(2\pi k) + O(1/k)$ , where  $h(k) = n \log k - k \log k + k$ . Its real maximum is at  $x$ , and  $h'' \leq -n/(4x^2)$  on  $[x/2, 2x]$ . Comparing with the term nearest  $x$  gives the bounds there, with only polynomial summation factors. Below  $x/2$  concavity gives an exponential loss. Above  $2x$ , the exact successive ratio  $(1 + 1/k)^n/(k+1)$  is at most  $e^{-cw}$ , giving a geometric tail.

Choose  $\varepsilon = w^{-3}$ . On the central rectangle,

$$\log R_n(KL) = \frac{n(n-1)}{2KL} + O\left(\frac{n^3}{(KL)^2}\right) = \frac{w^2}{2} + O(w^2\varepsilon + w^2/n + w^4/n) = \frac{w^2}{2} + o(1).$$

Three estimates control the complement, including rare small products.

1. If  $K, L \geq x/2$ , then  $\log R_n(KL) \leq 2w^2$ . Equation (14) bounds the noncentral contribution by  $\exp(2w^2 - cn/w^6)$ .
2. If  $\min(K, L) < x/2$  but  $KL \geq n^{3/2}$ , then  $\log R_n(KL) \leq \sqrt{n}/2$ , giving  $2\exp(\sqrt{n}/2 - cn)$ .
3. If  $KL < n^{3/2}$ , bound the unnormalized sum (5) directly by

$$\binom{KL + n - 1}{n} \leq [e(\sqrt{n} + 1)]^n.$$

Since  $e^{-2} \sum 1/(k!l!) \leq 1$ , this region has mass at most  $\exp(\frac{1}{2}n \log n + O(n))$ . The single Dobinski term nearest  $x$  gives

$$\log(B_n^2/n!) \geq n \log n - 2n \log w - n + 2n/w - O(\log n),$$

so this region is negligible even compared with  $B_n^2/n!$ .

We have proved  $Z_n \sim (B_n^2/n!)e^{w^2/2}$ . Together with (12) and the classical Bell equivalent below, this proves Theorem 1.1. The last two tail estimates remain exponentially small after multiplication by any fixed power of  $n$ ; we will use this fact again.

## 4 Bell expansions and uniform finite shifts

The Bell EGF is  $\exp(e^z - 1)$ . At its saddle  $we^w = n$ , the cumulants are  $\kappa_j = e^w T_j(w)$ , with  $T_j$  the Touchard polynomial. In particular,

$$\kappa_2 = n(w + 1), \quad \kappa_3 = n(w^2 + 3w + 1), \quad \kappa_4 = n(w^3 + 6w^2 + 7w + 1).$$

The classical saddle method [6] gives, to every fixed order, rational functions of  $w$  and controlled remainders. We record the precise form needed here:

$$B_n = \frac{\exp\{n(w - 1 + 1/w) - 1\}}{\sqrt{w + 1}} \left( 1 + \frac{\beta_1(w)}{n} + O\left(\frac{(1 + w)^2}{n^2}\right) \right), \quad (15)$$

where

$$\beta_1(w) = -\frac{w^2(2w^2 + 7w + 10)}{24(w + 1)^3}. \quad (16)$$

At order  $J$ , the remainder is  $O_J((1 + w)^{J+1}/n^{J+1})$ .

[Added 5 October 2026, batch 100: the order-one case of (15), with  $\beta_1 = -P(r)$  and the same error  $O(r^2/n^2)$ ,  $r = w$ , is already proved in the collection report `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277364-bell-asymptotics` (its `A277364_asymptotics.tex`, Section “The Bell saddle and its first correction”, equation `eq:bellfinal`), whose pre-correction `eq:precorrection` is the Gaussian correction displayed below and whose remark on two normalizations is the  $1/(12n)$  step below. The paragraph below is a second route to that statement; the all-orders form, its uniformity in  $w$  and the shifts (17) are not in that report. The all-orders uniformity is argued here in one paragraph and is recorded as Question I.5. ]

Here are direct uniformity details. On the circle  $z = we^{i\theta}$ , the logarithmic modulus loss is at least  $cnw\theta^2$  for  $|\theta| \leq c_0/w$ . Beyond this interval, discarding the inner cosine gives a loss at least  $cn/w^2$ . For the central variable  $X = \theta\sqrt{n(w + 1)}$ , put  $\epsilon = \sqrt{w/n}$ . The exact phase before subtracting its linear and quadratic terms is

$$\epsilon^{-2} \left[ \exp \left\{ w \left( \exp \frac{i\epsilon X}{\sqrt{w(w + 1)}} - 1 \right) \right\} - 1 \right].$$

Its remaining singularity at zero is removable, and its correction is  $O(\epsilon|X|^3)$  uniformly for  $w \geq 1$ . On the auxiliary circle  $|\epsilon| = c_J/(1 + |X|)^3$ , it is bounded. Cauchy expansion through odd degree  $2J+1$ , followed by symmetric Gaussian integration with cutoff  $|X| \leq \epsilon^{-1/4}$ , gives error  $O_J(\epsilon^{2J+2})$ . This proves the stated fixed-order remainder. The first Gaussian correction is

$$\frac{1}{n} \left\{ \frac{w^3 + 6w^2 + 7w + 1}{8(w+1)^2} - \frac{5(w^2 + 3w + 1)^2}{24(w+1)^3} \right\}.$$

Adding the factorial correction  $1/(12n)$  gives (16).

Let  $b(x)$  be the logarithm of the leading factor in (15). Exactly,

$$b'(x) = w_x - \frac{w_x}{2x(1+w_x)^2}, \quad w_x = W(x),$$

and  $b''(x) = w_x/(x(1+w_x)) + O(x^{-2})$ ,  $b'''(x) = O(x^{-2})$ . Uniformly for  $0 \leq s \leq 2D+2$ ,

$$\begin{aligned} \log \frac{B_{n-s}}{B_n} = & -sw + \frac{1}{n} \left\{ \frac{sw}{2(w+1)^2} + \frac{s^2w}{2(w+1)} \right\} \\ & + O\left( \frac{s^3 + s(1+w) + (1+w)^2}{n^2} \right). \end{aligned} \quad (17)$$

This follows by Taylor expansion of the explicit smooth Bell model, using its remainder at the two endpoints. No derivative of an uncontrolled remainder is taken. At every fixed order, the same argument gives coefficients polynomial in  $s$ , rational in  $w$ , with error  $n^{-J-1}$  times a fixed polynomial in  $s, w$ .

## 5 A finite marked profile expansion

We now prove Theorem 1.2, including the effects of all automorphisms. For a finite vector  $b = (b_2, b_3, \dots)$  with  $h = \sum b_i$ , define  $E(b)$  on a labeled multiset of cycles of these lengths by

$$E(b) = \sum_{\pi \in \Pi_h} \prod_{S \in \pi} \sum_{k \geq 2, k|\gcd(\ell_i: i \in S)} k^{|S|-1}, \quad E(0) = 1. \quad (18)$$

It is a finite integer. A recurrence chooses the block containing the least labeled selected cycle and applies the same rule to its complement. If  $m_i$  counts the cycles of length  $i$  of  $\sigma$ , the exact formula is

$$f(\sigma) = \sum_{0 \leq b_i \leq m_i, i \geq 2} E(b) \prod_i \binom{m_i}{b_i} B_{c-h}. \quad (19)$$

For  $f(\sigma)^2$ , take two independent selections  $b, b'$ . Their binomial factors multiply. This includes every overlap of the two selected sets: they need not be disjoint, and no additional matching factor is present.

### Two positive cutoffs

First restrict to  $d = n - c \leq D = A_J \log^2(n+1)$ ; equation (11) controls the discarded part. Let  $M$  be moved support. On this range  $M \leq 2D$ , and the normalized degree- $h$  sum in (19) is at most  $(\eta M)^h$ , where  $\eta = C' \log n/n$ . Terms with  $h + h' > J$  in its square have total at most

$$\sum_{j > J} (j+1)(\eta M)^j = O_J((\eta M)^{J+1}).$$

Summing positive weighted terms gives relative error  $O_J(n^{-J-1} \log^{3J+3} n)$ .

Next define the long-cycle defect

$$v = \sum_{i \geq 3} (i-2)m_i.$$

Under the probability law with weights  $B_{c(\sigma)}^2/(n!Z_n)$ , first sample positive  $K, L$  with weights proportional to  $(KL)^{\bar{n}}/(K!L!)$ , then sample an Ewens permutation with parameter  $T = KL$ . Its factorial moments are exactly

$$\mathbb{E} \prod_i (m_i)_{\underline{a_i}} = \frac{(n)_s T^{\sum_i a_i}}{\prod_i i^{a_i} (T+n-s)^{\bar{s}}}, \quad s = \sum_i i a_i \leq n. \quad (20)$$

The formula follows by marking the ordered cycles and summing over the remaining permutation. The region  $\min(K, L) < x/2$  is negligible by the weighted-mass tails. Otherwise  $T \geq n^2/(4w^2)$ . For  $u \geq 1$ , expansion in the nonnegative quantities  $u^{i-2} - 1$  and (20) give

$$\mathbb{E}(u^v \mid T) \leq \exp \left\{ \sum_{j \geq 1} \frac{n^{j+2}}{(j+2)T^{j+1}} (u^j - 1) \right\}.$$

Take  $u = n/(16w^4)$ . For large  $n$  it exceeds one, and

$$\sum_{j \geq 1} \frac{n^{j+2} u^j}{(j+2)T^{j+1}} \leq 4w^2 \sum_{j \geq 1} (4w^2)^{-j} \leq 2.$$

Therefore

$$\mathbb{P}(v > J \mid T) \leq e^2 (16w^4/n)^{J+1}. \quad (21)$$

On the retained deficit range  $f^2/B_c^2$  is uniformly bounded. Thus discarding  $v > J$  in the unweighted count has relative error  $O_J(n^{-J-1}w^{4J+4})$ , plus a negligible tail.

### The exact retained summand

Fix the finitely many long-cycle multiplicities  $(m_i)_{i \geq 3}$  with  $v \leq J$ , and write

$$H = \sum_{i \geq 3} m_i, \quad t = m_2, \quad d = t + v + H, \quad M = 2t + v + 2H.$$

Choose  $b, b'$  with  $h + h' \leq J$ . Their contribution to  $a_n/(B_n^2/n!)$  is

$$\begin{aligned} & \frac{(n)_M}{2^t t! \prod_{i \geq 3} i^{m_i} m_i!} \frac{B_{n-d-h} B_{n-d-h'}}{B_n^2} E(b) E(b') \\ & \cdot \binom{t}{b_2} \binom{t}{b'_2} \prod_{i \geq 3} \binom{m_i}{b_i} \binom{m_i}{b'_i}. \end{aligned} \quad (22)$$

After extracting  $(w/n)^{2d+h+h'}$  from the Bell factors and  $n^M$  from the falling factorial, its explicit scale is

$$\frac{\lambda^t}{t!} n^{-v-h-h'} \frac{w^{2v+2H+h+h'}}{\prod_{i \geq 3} i^{m_i} m_i!}, \quad \lambda = w^2/2, \quad (23)$$

times the finite selection factors in (22).

For a requested order, retain  $v + h + h' \leq J$ . The normalized falling factorial expands by

$$\log \frac{(n)_M}{n^M} = - \sum_{j \geq 1} \frac{n^{-j}}{j} \sum_{a=0}^{M-1} a^j.$$

Together with Section 4, every retained coefficient is polynomial in  $t$  and rational in  $w$ , and every remainder is  $n^{-J-1}$  times a fixed polynomial in  $t, w$ , uniformly for  $t \leq D$ . Profiles inadmissible at a small integer  $t$  have zero binomial selection factors; the finitely many small cases may equivalently be separated.

Extend each polynomial sum in  $t$  to infinity. Since  $D$  is a sufficiently large multiple of  $\log^2 n$ , the polynomial-weighted Poisson tail is  $\exp(-c \log^2 n)$ . Finally,

$$e^{-\lambda} \sum_{t \geq 0} \frac{t^k \lambda^t}{t!} = T_k(\lambda). \quad (24)$$

This produces  $Q_0, \dots, Q_J$  by finite rational operations. The same positive moments sum all error polynomials. Both cutoffs are already controlled, proving Theorem 1.2.

## 6 The first corrections and exact checks

The weighted count has the same algorithm with no selected cycles. Its first correction can also be checked through

$$s(n, n-d) = \frac{n^{2d}}{2^d d!} \left\{ 1 - \frac{d(2d+1)}{3n} + O\left(\frac{(d+1)^4}{n^2}\right) \right\}, \quad d \leq D.$$

The all-transposition term contributes  $-d(2d-1)/n$ ; one 3-cycle contributes  $4d(d-1)/(3n)$ . Profiles of long defect  $j \geq 2$  are bounded by the geometric series in  $8d^2/n$ , proving the uniform error. Combining with (17) and taking Poisson moments gives

$$R_1(w) = \frac{w^4(w-2)}{12(w+1)} - \frac{w^2}{2(w+1)^2}. \quad (25)$$

At extra order one, the only marked possibility is a single transposition, on either shore. Its weight is one and its selection factor is  $t$ . The extra Bell shift contributes  $w/n$ , so the Poisson mean gives  $2\lambda w/n = w^3/n$ . Thus  $Q_1 = R_1 + w^3$ , proving (3). The displayed fourth-degree errors average to  $O((1+w)^{10}/n^2)$ ; the two cutoff errors at this order are respectively  $O(w^6/n^2)$  and  $O(w^8/n^2)$ .

[Added 5 October 2026, batch 100: the marked term  $w^3/n$  is computed exactly, as a positive weighted count  $Y_{n-2}$  with  $Y_{n-2}/Z_n = w^3/n + O((1+w)^7/n^2)$ , in Part II (Lemmas 14.1 and 15.1), which also shows that it is the total weight of the single leaf transpositions. ]

The finite algorithm yields

$$Q_2(w) = \frac{w^3 \mathcal{P}(w)}{288(w+1)^5}, \quad (26)$$

where

$$\begin{aligned} \mathcal{P}(w) = & w^{10} + 23w^9 + 343w^8 + 1441w^7 + 2600w^6 + 2440w^5 \\ & + 1944w^4 + 1980w^3 + 1068w^2 - 324w - 576. \end{aligned}$$

The source code constructs the marked profiles, including independent overlapping selections, and uses exact rational Poisson moments. It explicitly rejects floating coefficients.

[Added 5 October 2026, batch 100:  $Q_2$  is printed without a derivation: it rests on the two delivered programs `code/01-asymptotics-compute_first_two.py` and `code/01-asymptotics-reconstruct_Q2.py`, which compute it by different compressions and agree (rerun on copies by the intake). The second records a decomposition of  $Q_2$  into seven contributions; the four without selected cycles add up to the second coefficient of the weighted count,

$$R_2(w) = \frac{w^3(w^{10} - w^9 + 7w^8 + w^7 - 40w^6 - 32w^5 - 216w^4 - 564w^3 - 420w^2 - 180w - 288)}{288(w+1)^5}.$$

As an independent partial check, the intake evaluated  $Z_n$  by the exact mass formula (5) to 40 significant digits and compared  $n^2\{Z_n/((B_n^2/n!)e^{w^2/2}) - 1 - R_1(w)/n\}$  with  $R_2(w)$ : the ratio is 1.0412, 1.0400, 1.0245, 1.0144, 1.0076, 1.0040 at  $n = 100, 300, 1000, 3000, 10^4, 3 \cdot 10^4$ , decreasing as a further relative correction of order  $\text{poly}(w)/n$  would. This is numerical evidence for the unmarked part of  $Q_2$  only; the three marked contributions were not checked independently, and a printed derivation of  $Q_2$  remains Question I.8. ]

A separate standard-library checker uses the exact fixed-partition recurrence and Burnside's sum. It checks 2,714 cycle types through  $n = 20$  and reproduces every listed sequence value there. Independent checks additionally enumerate fixed set partitions directly at small sizes and verify the mass normalization and Ewens factorial moments. These finite tests validate identities and transcription, not the asymptotic tail proofs.

## 7 Explicit inverse models and integer thresholds

Put

$$\Phi(x) = x(w - \log w - 1 + 2/w) + \frac{w^2}{2} - 2 - \log(w+1) - \frac{1}{2} \log(2\pi x), \quad w = W(x). \quad (27)$$

Its exact derivative is

$$\Phi'(x) = w - \log w + \frac{1}{x} \left( \frac{w^2}{w+1} - \frac{w}{(w+1)^2} - \frac{1}{2} \right) \sim \log x. \quad (28)$$

In particular the leading model is eventually increasing. If  $\nu(y)$  is the eventual solution of  $\Phi(\nu) = \log y$ , Theorem 1.1 gives

$$\nu(a_n) = n + o(1/\log n).$$

Nearest-integer rounding therefore recovers the index on the sequence range eventually. Also  $a_{n+1}/a_n \sim n/W(n)^2$ , so the actual sequence is eventually strictly increasing.

Combining the all-orders Bell expansion and Stirling with (2) gives rational functions  $P_j(w)$  such that

$$a_n = e^{\Phi(n)} \left\{ \sum_{j=0}^J \frac{P_j(W(n))}{n^j} + O_J(n^{-J-1}(1+W(n))^{M'_J}) \right\}, \quad P_0 = 1. \quad (28a)$$

Their finite rule multiplies the  $Q$  series by the square of the Bell correction series and the reciprocal Stirling series. For example,

$$P_1(w) = \frac{w^7 + 12w^6 + 33w^5 + 32w^4 - 2w^3 - 19w^2 - 3w - 1}{12(w+1)^3}. \quad (29)$$

This equals  $Q_1 + 2\beta_1 - 1/12$ . For fixed  $J$  define the explicit smooth tail model

$$F_J(x) = e^{\Phi(x)} \sum_{j=0}^J P_j(W(x)) x^{-j}.$$

It is positive and strictly increasing eventually, with logarithmic derivative asymptotic to  $\log x$ . Write  $x_J(y)$  for its tail inverse. For the discrete threshold  $N(y) = \min\{n : a_n \geq y\}$ , there is a constant  $C_J$  such that

$$\lceil x_J(y) - \varepsilon_J(y) \rceil \leq N(y) \leq \lceil x_J(y) + \varepsilon_J(y) \rceil, \quad \varepsilon_J(y) = \frac{C_J x_J^{-J-1} (1+W(x_J))^{M'_J}}{\log x_J} \quad (30)$$

for sufficiently large  $y$ , enlarging  $C_J$  at endpoints if necessary. This follows by the mean-value theorem and monotone bracketing. In particular a single ceiling is valid when both endpoint ceilings agree, and range inversion has vanishing index error.

For a compositional rule, let  $X = \nu(y)$  and formally put  $h(X) = \log(\sum_{j \geq 0} P_j(W(X))X^{-j})$ . Then the compatible inverse expansion is

$$X + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left( \frac{1}{\Phi'(X)} \frac{d}{dX} \right)^{k-1} \frac{h(X)^k}{\Phi'(X)}. \quad (31)$$

At every fixed order, only finitely many terms are needed. Taylor's theorem applies to the finite smooth model, and (30) transfers the error to thresholds. No exact interpolation of the discrete sequence or unconditional ceiling formula is imposed.

## 8 A consequence for recurrence equations

**Proposition 8.1.** *The sequence  $a_n$  is not P-recursive, even with complex polynomial coefficients. Equivalently, its ordinary generating function is not D-finite.*

*Proof.* Theorem 1.1 gives

$$\log \frac{n!}{a_n} = 2n \log \log n + O(n).$$

Thus  $E(z) = \sum_{n \geq 0} (a_n/n!)z^n$  is entire. If  $a_n$  were P-recursive, substitution of  $a_{n+j} = (n+j)![z^{n+j}]E(z)$  in its polynomial recurrence would make the coefficients of  $E$  P-recursive, so  $E$  would satisfy a nonzero linear differential equation with polynomial coefficients.

Any entire solution of such an equation has a bound  $\max_{|z|=r} |E(z)| \leq \exp(Cr^M)$  for some fixed  $C, M > 0$  and large  $r$ . Indeed, outside a circle containing the zeros of the leading polynomial, the companion first-order system has rational coefficients bounded by a fixed power of  $r$ . Integrating along each radial segment and applying Gronwall gives the stated bound uniformly in the direction; the initial values on the fixed circle are bounded. Cauchy's coefficient estimate at  $r = (n/(CM))^{1/M}$  would then imply

$$\log \frac{n!}{a_n} \geq \frac{n}{M} \log n - O(n),$$

contradicting the displayed coefficient growth. This proves the claim.  $\square$

The ODE criterion is classical; this is a direct consequence of the proved relative asymptotic, with no separate priority assertion.

## 9 More than two distinguished shores

Let  $a_n^{(r)}$  count simultaneous  $S_n$  orbits on ordered  $r$ -tuples of set partitions of  $[n]$ , equivalently  $r$ -partite multihypergraphs with  $n$  hyperedges, distinguished shores and no isolated vertices.

**Proposition 9.1.** *For every fixed integer  $r \geq 3$ ,*

$$a_n^{(r)} = \frac{B_n^r}{n!} \left\{ 1 + O_r \left( \frac{\log^r n}{n^{r-2}} \right) \right\}.$$

*More precisely, the first relative excess is  $W(n)^r / (2n^{r-2})(1 + o(1))$ .*

*Proof.* Burnside gives  $a_n^{(r)} = \sum_{\sigma} f(\sigma)^r / n!$ . Equations (8), (9) and (10) bound the normalized deficit- $d$  contribution by

$$\frac{(2C \log(n+1))^{rd}}{d! (n)_d^{r-2}}.$$

The geometric mean of the largest  $d$  factors of  $n!$  is at least  $(n!)^{1/n} \geq n/e$ , so  $(n)_d \geq (n/e)^d$ . Summing  $d \geq 1$  gives  $\exp(O_r(\log^r n / n^{r-2})) - 1$ . For  $d = 1$ , a transposition fixes  $B_{n-1} + B_{n-2}$  set partitions, so its contribution is  $\binom{n}{2} [(B_{n-1} + B_{n-2}) / B_n]^r \sim W(n)^r / (2n^{r-2})$ . The bound just proved makes the  $d \geq 2$  contribution of smaller order.  $\square$

Thus two shores form an exceptional fixed-shore case: their correction factor diverges, whereas for every fixed larger number of shores the identity term is a relative equivalent.

## 10 Further questions

The proof suggests the following extensions, which are not asserted here.

- I.1. Effective remainder constants and useful numerical onset bounds for coefficient estimates and inverse rounding.
- I.2. A scaled distributional description of rare nontrivial vertex automorphisms, beyond their total weighted contribution.

[Added 5 October 2026, batch 100: answered at leading order by Part II. For the uniform measure on isomorphism classes, Theorem 12.1 gives  $\mathbb{P}(\text{Aut}(G) \neq 1) = 2w^3/n + O((1+w)^7/n^2)$ , and shows that every group other than the trivial group and a single leaf-transposition  $C_2$  has probability  $O((1+w)^6/n^2)$ ; Lemma 15.1 identifies the  $w^3$  of  $Q_1 = R_1 + w^3$  as the exact leaf-transposition mass. The question stays open, re-scoped: the next coefficient, at order  $n^{-2}$ , of the symmetry probability, and the law of the group on the exceptional graphs of probability  $O((1+w)^6/n^2)$ ; this is Part II's Question II.1. ]

- I.3. Uniform all-orders multipartite expansions, including a regime where the number of distinguished shores grows with  $n$ .
- I.4. Precisely normalized exponentially small sectors after optimal algebraic truncation, with their dependence on any chosen interpolation made explicit.

[Added 5 October 2026, batch 100: under the standing rule for unproved claims, the following steps of Part I are sketched rather than written out in full. None is contradicted by the intake's checks, and none is used here as more than its source proves. ]

- I.5. *Uniformity of the all-orders Bell saddle* (Section 4). The remainder  $O_J((1+w)^{J+1}/n^{J+1})$  of (15) at order  $J$ , uniform in  $w$ , is argued in one paragraph: a removable singularity, an auxiliary circle  $|\epsilon| = c_J/(1+|X|)^3$ , Cauchy expansion through odd degree  $2J+1$  and Gaussian integration with cutoff  $|X| \leq \epsilon^{-1/4}$ . The order-one case is also proved in the collection report `a277364-bell-asymptotics` (note in Section 4). Missing: a written proof at every order, with the tail of the Gaussian integral and the dependence of the constants on  $J$  made explicit.
- I.6. *The exponents  $M_J$  and  $M'_J$*  (Theorem 1.2 and (28a)). Section 5 shows that every remainder is  $n^{-J-1}$  times a fixed polynomial in  $t$  and  $w$ , so that some finite exponent exists; no value is computed for  $J \geq 2$  (the two cutoffs contribute  $\log^{3J+3} n$  and  $w^{4J+4}$ , the Taylor remainders an unspecified degree). Part II's exponents (Question II.4) inherit this. Missing: explicit  $M_J, M'_J$ .

- I.7. *The order-one remainder*  $(1 + w)^{10}$  (Theorem 1.2). It is asserted from the three error sizes displayed in Section 6 (“the displayed fourth-degree errors average to  $O((1 + w)^{10}/n^2)$ ”); the averaging over the Poisson law is not written out. Since  $Q_2(w) \sim w^8/288$ , an exponent 8 would match the size of the next term; whether  $(1 + w)^8$  suffices is open.
- I.8. *A printed derivation of  $Q_2$*  ((26)). The coefficient rests on the two delivered programs only; the intake’s independent check covers its unmarked part (note after (26)). Missing: a derivation in the text, in particular of the contributions with one or two selected cycles.

## 11 Source scope and reproducibility

The current OEIS entry gives a cycle-index formula but no asymptotic equivalent. The exact-sequence theorem located in the bounded primary-literature search is the logarithmic result of [3]. The two incidence-matrix papers [4, 5] explicitly use zero-one entries. Their fully unlabeled shore-preserving class is A049311, and their main relative formula uses labeled rows and columns. It cannot be transferred to A007716 without a new argument. Bell saddles, Burnside averaging, Dobinski’s identity and Poisson moment expansions are classical. We claim no exhaustive determination of priority.

The archive contains editable source, exact code and output, independent checks and a manifest. The mathematical statements are ordinary unrefereed proofs with exact regression tests; they are not Lean formalizations. The all-orders assertions concern fixed algebraic orders with logarithmic coefficient functions. They do not classify all exponentially small sectors or Stokes data, and no effective numerical onset is supplied.

[*Added 5 October 2026, batch 100:* in this repository the delivered package is shipped without its PDF, its checksum manifest `SHA256SUMS` and its `build_local.sh` (a byte copy of a script already in the repository); its programs are in `code/` and their recorded outputs in `data/`, with the prefix `01-asymptotics-`, and its verification note is `01-asymptotics-verification.md`. The report’s `README.md` maps the delivered names to the shipped ones and gives rerun instructions that work on a copy (each program writes its output beside itself). The intake reran all five programs on copies on 5 October 2026; every output equals the recorded one apart from line endings. The neighbouring collection reports are named in Appendix A. ]

## Part II

# Rare symmetries and connectivity (Report 92)

[*Added 5 October 2026, batch 100:* Part II is manuscript 02 of batch 100 (Research Report 92, 2 October 2026), printed whole except for its Section 2, whose restatements of Part I are replaced by references where Part I prints the same display (Section 13). Its symbols are renamed as in Table 1; “the preceding report” is Part I. Its section and theorem numbers are those of manuscript 02 plus eleven (its Theorem 1.1 is Theorem 12.1 here). ]

## 12 The model and the principal results

Let  $\mathcal{G}_n$  be the finite set of isomorphism classes of bipartite multigraphs with  $n$  edges, no isolated vertices and two distinguished shores. Isomorphisms preserve the shores; transposition is not an additional equivalence. Write  $a_n = |\mathcal{G}_n|$ , with  $a_0 = 1$ . Equivalently,  $a_n$  counts nonnegative integer matrices of total sum  $n$ , up to separate row and column permutations, after zero rows and columns are removed. This is A007716 [1].

Let  $c_n^\circ$  count its connected members for  $n \geq 1$  and set  $c_0^\circ = 0$ . Unique decomposition into connected components gives the formal identity

$$\mathcal{A}(z) := \sum_{n \geq 0} a_n z^n = \prod_{m \geq 1} (1 - z^m)^{-c_m^\circ}. \quad (32)$$

Thus  $c_n^\circ$  agrees with A007718 at every positive index [2]; that OEIS entry separately assigns the conventional value 1 at index zero. The first positive-index connected counts are

$$1, 3, 6, 17, 40, 125, 354, 1159, 3774, 13113, \dots$$

All probability statements below use the uniform measure on  $\mathcal{G}_n$ , not a vertex-labeled or automorphism-weighted measure.

A *leaf transposition* interchanges two degree-one vertices on the same shore that have a common neighbor and fixes every other vertex. Degree counts edge multiplicity. Let  $\Lambda(G)$  be the number of these individual transpositions, and write  $w = W(n)$  for the positive Lambert function,  $we^w = n$ .

**Theorem 12.1.** *For a uniformly chosen  $G \in \mathcal{G}_n$ ,*

$$\mathbb{P}(\text{Aut}(G) \neq 1) = \frac{2w^3}{n} + O\left(\frac{(1+w)^7}{n^2}\right). \quad (33)$$

Moreover,

$$\mathbb{P}(\text{Aut}(G) \text{ is neither trivial nor generated by one leaf transposition}) = O\left(\frac{(1+w)^6}{n^2}\right). \quad (34)$$

Here  $\text{Aut}(G)$  always denotes the shore-preserving vertex group.

In particular, conditional on having a nontrivial vertex automorphism, the group is a single leaf-transposition  $C_2$  with probability  $1 - O((1+w)^3/n)$ . The two possible shores occur equally often by transposing the entire graph.

**Theorem 12.2.** *For each fixed integer  $J \geq 0$ , define integers  $\rho_j$  by the formal reciprocal*

$$\sum_{j \geq 0} \rho_j z^j = \frac{1}{\mathcal{A}(z)}, \quad (\rho_0, \rho_1, \rho_2, \rho_3, \rho_4) = (1, -1, -3, -3, -8). \quad (35)$$

Then

$$c_n^\circ = \sum_{j=0}^J \rho_j a_{n-j} + O_J(a_{n-J-1}). \quad (36)$$

Consequently,

$$\mathbb{P}(G \text{ disconnected}) = \frac{w^2}{n} + \frac{w^3(3w^3 + 5w^2 + 3w + 2)}{(w+1)^2 n^2} + O\left(\frac{(1+w)^{\xi_1}}{n^3}\right) \quad (37)$$

for some fixed finite  $\xi_1$ . An isolated-edge component is present with exact probability  $a_{n-1}/a_n$ , and disconnection without such a component has probability  $O(w^4/n^2)$ .

The reciprocal in (35) is purely formal. Neither it nor the product (32) is asserted to converge for nonzero  $z$ .

## 13 Inputs from Part I

[Added 5 October 2026, batch 100: manuscript 02's Section 2, "Inputs from the preceding asymptotic analysis", restated the results of Report 88 that it uses "so that the dependence is explicit". Report 88 is Part I. Its displays (Z), (allinput), (P1), (burn), (ewens) and (bellshift) are Part I's (4) with (5) and (8), (28a), (29), (6), (20) and (17), and are not printed again; its displays (inputs), (markbound) and (goodT), whose form is not displayed in Part I, are kept with notes. Its prose is kept. ]

Part I proves the following results. We state the exact forms used so that the dependence is explicit. Let  $B_n$  be the Bell number. The weighted count  $Z_n = \sum_{G \in \mathcal{G}_n} 1/|\text{Aut}(G)|$  has the three forms (4), (5) and the second formula of (8), where  $c(\sigma)$  is the number of element cycles. The degree-zero binomial expression means its polynomial value 1.

The established relative estimates are

$$\begin{aligned} a_n &\sim Z_n \sim \frac{B_n^2}{n!} e^{w^2/2} \sim e^{\Phi(n)}, \\ Z_n &= \frac{B_n^2}{n!} e^{w^2/2} \{1 + O((1+w)^4/n)\}, \end{aligned} \quad (38)$$

with  $\Phi$  as in (27).

[Added 5 October 2026, batch 100: the first line is Theorem 1.1 with (12);  $e^{\Phi(n)}$  is the right-hand side of (1). The second line is not displayed in Part I. It follows from Part I's Section 6, where the weighted count is expanded by the same algorithm with no selected cycles and has first coefficient  $R_1(w) = O(w^4)$ , (25): the order-one remainder  $O((1+w)^M/n^2)$  of that expansion is  $O((1+w)^4/n)$  for every fixed  $M$ , because  $(1+w)^{M-4} \leq n$  eventually. ]

For every fixed  $J$  there are rational functions  $P_j$ ,  $P_0 = 1$ , and a finite exponent  $M'_J$  such that (28a) holds; the first coefficient  $P_1$  is (29). The coefficients have a finite marked-cycle and Poisson-moment construction; no infinite formal expansion is used as a convergent equality.

For clarity we also recall the positive cutoff and moment facts used below. A pair of set partitions of  $[n]$  represents a graph with edge labels. If  $f(\sigma)$  is the number of invariant set partitions, then  $a_n$  is given by Burnside's average (6). The expansion of  $f$  marks the element cycles used in nontrivial orbits of vertex blocks. Its unmarked term is  $B_c$ . A single selected cycle of length  $i$  has weight  $\tau(i) - 1$ , where  $\tau(i)$  counts divisors. Selecting a transposition has weight one and replaces  $B_c$  by  $B_{c-1}$ .

[Added 5 October 2026, batch 100: the weight  $\tau(i) - 1$  is  $E(b)$  of (18) for a single selected cycle of length  $i$  ( $h = 1$ , so  $|S| = 1$  and every divisor  $k \geq 2$  of  $i$  contributes  $k^0 = 1$ ), and (19) is the expansion meant. ]

Write  $d = n - c$  and take  $D = A \log^2(n+1)$  with sufficiently large fixed  $A$ . The part  $d > D$  of (6), relative to  $B_n^2/n!$ , is  $O(\exp(-\gamma \log^2 n))$ ;  $A$  can be increased to obtain any fixed required  $\gamma$ . On  $d \leq D$ ,

$$\frac{B_{c-h}}{B_c} \leq \eta^h, \quad \eta = Cw/n, \quad \text{total normalized degree-}h \text{ marked mass} \leq (Cdw/n)^h. \quad (39)$$

The second bound follows by selecting the nontrivial cycles, whose total moved support is at most  $2d$ , and using  $B_h \leq h!$  for their grouping. In particular the sum of all marked degrees is bounded uniformly on this range.

[Added 5 October 2026, batch 100: Part I proves the first bound with  $\eta = C' \log n/n$  (Section 2, "A quantitative comparison with the weighted count", and Section 5). Since  $w = \log n - \log \log n + o(1)$ , the two forms are equivalent up to the constant; this is a restatement, not a discrepancy. ]

Under the probability measure on  $S_n$  proportional to  $B_c^2$ , one can first choose positive  $K, L$  with weight proportional to  $(KL)^{\bar{n}}/(K!L!)$ , then an Ewens permutation of parameter  $T = KL$ . For nonnegative integers  $a_i$ , with  $s = \sum ia_i \leq n$ , its factorial moments given  $T$  are (20), where  $m_i$  is the number of  $i$ -cycles. The region

$$\min(K, L) < n/(2w) \quad (40)$$

has probability smaller than every fixed inverse power of  $n$  under this weighted law. Part I proves this directly, including the rare small-product region; it is not inferred from an unweighted concentration estimate.

[Added 5 October 2026, batch 100:  $n/(2w) = x/2$ , so (40) is the region that Part I discards in Section 5 (“The region  $\min(K, L) < x/2$  is negligible by the weighted-mass tails”), using the three complement estimates of Section 3. Manuscript 02 writes the Ewens exponents as  $r_i$ ; they are Part I’s  $a_i$ . ]

Finally its uniform Bell shifts, for  $s = O(\log^2 n)$ , are (17), proved there uniformly for  $0 \leq s \leq 2D + 2$ . All these bounds concern the already proved estimates of Part I; the new work below converts them into uniform-isomorphism probabilities and connected counts.

## 14 The exact leaf transposition mass

Define

$$Y_m = \frac{1}{2} \sum_{\Gamma \in \mathcal{G}_m} \frac{|V(\Gamma)| + 2}{|\text{Aut}(\Gamma)|}, \quad (41)$$

where  $|V(\Gamma)|$  counts all vertices. A marked neighbor may be an existing vertex of  $\Gamma$  or a new isolated vertex before the two leaves are attached. The factor  $1/2$  accounts for the unordered pair of added leaves. Sum over the two shores to obtain (41).

There are useful exact alternatives. Introducing row and column vertex-count variables into (5), or differentiating the operation that adds isolated vertices, gives

$$Y_m = \frac{e^{-2}}{2} \sum_{k, l \geq 0} \frac{(k+l) \binom{kl+m-1}{m}}{k!l!} = \frac{1}{m!} \sum_{\sigma \in S_m} B_{c(\sigma)} B_{c(\sigma)+1}. \quad (42)$$

The last equality uses Dobinski’s identity and the rising-factorial cycle identity. It also holds at  $m = 0$ , where  $Y_0 = 1$ .

**Lemma 14.1.** *For  $n \geq 2$ , the marked Burnside term in which exactly one transposition is selected on exactly one shore is*

$$Y_{n-2} = \frac{1}{n!} \sum_{\sigma \in S_n} 2m_2(\sigma) B_{c(\sigma)} B_{c(\sigma)-1} = \sum_{G \in \mathcal{G}_n} \frac{\Lambda(G)}{|\text{Aut}(G)|}. \quad (43)$$

*Proof.* For a cycle type  $\mu$ , put  $z_\mu = \prod_i i^{m_i} m_i!$ . Deleting the selected 2-cycle gives a type  $\nu$  of  $n - 2$  with  $c(\mu) = c(\nu) + 1$  and  $2m_2(\mu)/z_\mu = 1/z_\nu$ . Summing proves the first equality from (42).

To verify the group interpretation, fix an edge-labeled partition-pair representation of  $G$ . Its stabilizer in  $S_n$  maps onto  $\text{Aut}(G)$ . Its kernel has size

$$\Omega_G = \prod_{u,v} m_{uv}!,$$

because it permutes only parallel edges. Every vertex automorphism lifts, and each fiber has exactly  $\Omega_G$  elements. A leaf transposition has one forced 2-cycle on the two leaf edges. Its other edge cycles lie within parallel bundles, so they fix all remaining vertex blocks. In the marked expansion it therefore has exactly the prescribed single selected transposition on one shore. Conversely such a marking makes precisely two singleton vertex blocks exchange on one shore, while their edges lie in a fixed common block on the other shore. It gives one leaf transposition. There are  $n!/(\Omega_G |\text{Aut}(G)|)$  partition-pair realizations, and  $\Omega_G \Lambda(G)$  marked lifts for each realization. Division by  $n!$  yields  $\Lambda(G)/|\text{Aut}(G)|$ .  $\square$

Parallel-edge permutations can be numerous. The kernel cancellation in this proof is essential; they are not the vertex automorphisms in Theorem 12.1.

## 15 The residual and the symmetry probability

**Lemma 15.1.** *With  $Y_{n-2}$  as above,*

$$0 \leq \mathcal{R}_n := a_n - Z_n - Y_{n-2} = O\left(Z_n \frac{(1+w)^6}{n^2}\right), \quad \frac{Y_{n-2}}{Z_n} = \frac{w^3}{n} + O\left(\frac{(1+w)^7}{n^2}\right). \quad (44)$$

*Proof.* All marked terms are nonnegative. Remove the unmarked pair and the one-selected-transposition terms. On  $d \leq D$ , the total contribution having at least two selected cycles across the two shores is, by (39),

$$\sum_{j \geq 2} (j+1)(Cdw/n)^j = O(w^6/n^2)$$

relative to  $B_c^2$ . The remaining residual terms select a single cycle of length  $i \geq 3$ . Their relative contribution is at most

$$2\eta \sum_{i \geq 3} (\tau(i) - 1)m_i \leq 2\eta \sum_{i \geq 3} im_i.$$

On the complement of (40),  $T \geq n^2/(4w^2)$ ; equation (20) gives

$$\mathbb{E}\left(\sum_{i \geq 3} im_i \middle| T\right) \leq \sum_{i \geq 3} \frac{n^i}{T^{i-1}} = O(w^4/n). \quad (45)$$

The bad- $T$  part is negligible, since the sum is at most  $n$ . Thus one selected long cycle costs  $O(w^5/n^2)$ . The discarded  $d > D$  tail is also negligible. This proves the first estimate.

For the second estimate first retain only permutations consisting of fixed points and  $t$  transpositions, with  $t \leq D$ . Put  $\lambda = w^2/2$ . Their contribution to  $Y_{n-2}/(B_n^2/n!)$  is

$$\sum_{t \leq D} \frac{(n)_{2t}}{2^t t!} \frac{2t B_{n-t} B_{n-t-1}}{B_n^2} = \sum_{t \leq D} \frac{\lambda^t}{t!} \frac{2tw}{n} \left\{ 1 + O\left(\frac{(t+1)^2}{n}\right) \right\}. \quad (46)$$

Indeed (17) and the Taylor expansion of the falling factorial show that the logarithm of the remaining factor is  $O((t+1)^2/n)$  uniformly here. Its higher-order remainders are smaller on  $t = O(w^2)$ . Extending the Poisson sum past  $D$  has superpolynomially small error after  $A$  is chosen large. Since  $\mathbb{E}t = \lambda$  and  $\mathbb{E}[t(t+1)^2] = O((1+\lambda)^3)$ , equation (46) divided by  $e^\lambda$  is

$$\frac{w^3}{n} + O((1+w)^7/n^2).$$

The terms with a long cycle contribute at most  $2\eta \mathbb{E}[m_2 \mathbf{1}_{\{\exists i \geq 3: m_i > 0\}}]$  relative to  $Z_n$ , apart from negligible tails. By the mixed factorial moments in (20), on the good- $T$  region this is bounded using

$$\mathbb{E}\left(m_2 \sum_{i \geq 3} m_i \middle| T\right) \leq \frac{n^2}{2T} \sum_{i \geq 3} \frac{n^i}{iT^{i-1}} = O(w^6/n). \quad (47)$$

Terms whose total selected support exceeds  $n$  have zero moment and may simply be included in the upper bound. The bad- $T$  part is negligible since the random variable is at most  $n^2$ . Thus the extra marked mass is  $O((1+w)^7/n^2)$  relative to  $Z_n$ . Finally use the second line of (38); changing the normalization introduces an error of that same order.  $\square$

*Proof of Theorem 12.1.* Put  $g = |\text{Aut}(G)|$ . The sign of a vertex permutation is a homomorphism to  $\{1, -1\}$ . If  $\Lambda(G) > 0$  it is onto, and every counted leaf transposition lies in its odd coset. Therefore  $\Lambda(G) \leq g/2$ ; the inequality is trivial if  $\Lambda = 0$ . Consequently

$$\varrho(G) := \frac{g - 1 - \Lambda(G)}{g} \geq 0.$$

It vanishes for the identity group and for a group generated by one leaf transposition. For  $g \geq 3$ ,  $\varrho(G) \geq 1/2 - 1/g \geq 1/6$ . For  $g = 2, \Lambda = 0$ , it is  $1/2$ . These cases exhaust all other groups. Lemma 14.1 implies  $\mathcal{R}_n = \sum_G \varrho(G)$ , so the number of exceptional graphs is at most  $6\mathcal{R}_n$ .

For every nonexceptional graph,

$$\mathbf{1}_{\{\text{Aut}(G) \neq 1\}} = \frac{2\Lambda(G)}{|\text{Aut}(G)|}.$$

The two sides differ by at most one on an exceptional graph, since  $2\Lambda/g \leq 1$ . Lemma 15.1 now gives

$$\mathbb{P}(\text{Aut}(G) \neq 1) = \frac{2Y_{n-2}}{a_n} + O((1+w)^6/n^2).$$

Also  $a_n/Z_n = 1 + Y_{n-2}/Z_n + \mathcal{R}_n/Z_n = 1 + O((1+w)^3/n)$ . Substituting the second estimate in (44) proves (33) and the first proves (34).  $\square$

[Added 5 October 2026, batch 100: the identity  $\mathcal{R}_n = \sum_G \varrho(G)$  is the combination of  $a_n = \sum_G 1$ , (4) and Lemma 14.1:  $\sum_G (1 - 1/g - \Lambda(G)/g) = a_n - Z_n - Y_{n-2}$ . Together with Lemma 15.1 it shows that Part I's correction  $Q_1 = R_1 + w^3$  splits as the weighted count's  $R_1$  plus exactly the leaf-transposition mass, the residual entering only at order  $n^{-2}$ .]

## 16 Endpoint convolution and the giant component

The argument for connectivity needs more than  $a_{n-1}/a_n \rightarrow 0$ . In particular one must control graphs whose components are all small.

**Lemma 16.1.** *For each fixed positive integer  $q$ ,*

$$\sum_{q \leq k \leq n/2} a_k a_{n-k} = O_q(a_{n-q}). \quad (48)$$

*There is a constant  $\gamma > 0$  such that*

$$\sum_{n/4 \leq k \leq n/2} a_k a_{n-k} = O(a_n e^{-\gamma n}). \quad (49)$$

*Proof.* Direct differentiation of the explicit  $\Phi$  gives

$$\begin{aligned} \Phi'(x) &= \log x - 2 \log W(x) + \frac{1}{x} \left\{ \frac{W(x)^2}{1+W(x)} - \frac{W(x)}{(1+W(x))^2} - \frac{1}{2} \right\}, \\ \Phi''(x) &\sim 1/x. \end{aligned} \quad (50)$$

Choose a fixed integer  $n_0$  beyond the onset of  $\Phi'' > 0$  and beyond a two-sided constant comparison  $a_k \asymp e^{\Phi(k)}$ . No convexity of the actual sequence, or fixed numerical onset such as 4, is assumed.

Set  $\Psi_n(t) = \Phi(t) + \Phi(n-t)$ . For  $n_0 \leq t \leq n/4$ ,

$$\Psi'_n(t) \leq \Phi'(n/4) - \Phi'(3n/4) \longrightarrow -\log 3.$$

Thus the exponential terms on this interval decrease at least geometrically and their sum is bounded by a constant times their left endpoint. On  $n/4 \leq t \leq n/2$ ,  $\Psi_n$  is decreasing. Moreover,

$$\Phi(n/4) + \Phi(3n/4) - \Phi(n) = -\text{Ent}(1/4)n + o(n),$$

where  $\text{Ent}(p) = -p \log p - (1-p) \log(1-p) > 0$ . This follows immediately from  $\Phi(x) = x \log x - 2x \log \log x - x + o(x)$  at the fixed scaled arguments. The polynomial number of central terms is absorbed by a smaller positive  $\gamma$ , proving (49).

If  $q \geq n_0$ , the first interval starts at  $q$  and is  $O_q(a_{n-q})$ . If  $q < n_0$ , keep the finitely many  $k = q, \dots, n_0 - 1$  separately and start the geometric argument at  $n_0$ . The fixed-shift ratios from (28a) give

$$\frac{a_{n-j}}{a_n} \sim \left(\frac{w^2}{n}\right)^j \quad \text{for every fixed } j \geq 0, \quad (51)$$

so these finite contributions and the endpoint at  $n_0$  are  $O_q(a_{n-q})$ . The central part is smaller still. This proves (48).  $\square$

[Added 5 October 2026, batch 100: since  $\log x = W(x) + \log W(x)$ , the first line of (50) is Part I's (28); the asymptotic  $\Phi''(x) \sim 1/x$  is new here. ]

Call a component with more than  $n/2$  edges a *giant*; it is unique if present. Let  $e_n$  count graphs with no such component. A graph counted by  $e_n$  has a union of components with total edge count in  $[n/4, n/2]$ . Indeed choose a component already in that interval; if there is none, all components have fewer than  $n/4$  edges, and greedily accumulating them until first reaching  $n/4$  produces a union smaller than  $n/2$ . Hence

$$e_n \leq \sum_{n/4 \leq k \leq n/2} a_k a_{n-k} = O(a_n e^{-\gamma n}). \quad (52)$$

The bound counts possible pairs of unions and may overcount graphs, which is harmless. For all other graphs the giant and the remaining graph determine one another uniquely, giving the exact identity

$$a_n = \sum_{0 \leq j < n/2} a_j c_{n-j}^\circ + e_n. \quad (53)$$

*Proof of the finite-shift part of Theorem 12.2.* Since  $0 \leq c_k^\circ \leq a_k$ , truncate (53) at fixed  $J$ . Equations (48) and (52) give

$$a_n = \sum_{j=0}^J a_j c_{n-j}^\circ + O_J(a_{n-J-1}).$$

Apply this relation at  $n-i$  to order  $J-i$ , multiply by  $\rho_i$ , and sum over  $0 \leq i \leq J$ . The coefficient of  $c_{n-k}^\circ$  is  $\sum_{i=0}^k \rho_i a_{k-i} = \mathbf{1}_{\{k=0\}}$ . Every error has size  $O_J(a_{n-J-1})$ . This proves (36).  $\square$

## 17 Disconnection and finite fragments

Removal of one isolated-edge component is a bijection from graphs that have at least one such component to  $\mathcal{G}_{n-1}$ . Components of this type are indistinguishable, so no multiplicity factor appears. For a disconnected graph without an isolated-edge component, its smallest component has edge count between 2 and  $n/2$ . Selecting it and its complement, and overcounting if necessary, gives a bound by (48) with  $q = 2$ . Therefore

$$\mathbb{P}(G \text{ disconnected}) = \frac{a_{n-1}}{a_n} + O\left(\frac{a_{n-2}}{a_n}\right), \quad \mathbb{P}(\text{disconnected, no isolated edge}) = O(w^4/n^2). \quad (54)$$

Conditional on disconnection, the graph consists of a connected  $(n-1)$ -edge graph and a single isolated edge with probability  $1 - O(w^2/n)$ . One can see this either from (53) at fragment size one or from  $c_{n-1}^\circ/a_{n-1} = 1 - O(w^2/n)$ .

For the second correction, write

$$\alpha_1(w) = \frac{w^2}{w+1} - \frac{w}{(w+1)^2} - \frac{1}{2}.$$

Taylor expansion of the finite smooth model in (28a), rather than differentiation of its error, yields

$$\frac{a_{n-1}}{a_n} = \frac{w^2}{n} \left\{ 1 + \frac{\delta(w)}{n} + O\left(\frac{(1+w)^{\xi_2}}{n^2}\right) \right\}, \quad \delta(w) = \frac{w(2-w^2)}{(w+1)^2}. \quad (55)$$

Indeed the correction in the logarithmic ratio is  $-\alpha_1(w) + (w-1)/(2(w+1)) = \delta(w)$ , by (50). The factor  $1 + P_1(w)/n + \dots$  changes this ratio only at relative order  $n^{-2}$  times a fixed polynomial in  $w$ . Taking  $J = 2$  in (36), using  $a_{n-2}/a_n = w^4/n^2 + O(n^{-3}(1+w)^{\xi_3})$ , and combining  $w^2\delta(w) + 3w^4$  proves (37).

More generally, fix an isomorphism class  $\Gamma \in \mathcal{G}_j$ . For  $n > 2j$ , the event that the complement of the giant is exactly  $\Gamma$  has the exact probability

$$\mathbb{P}(\text{fragment} = \Gamma) = \frac{c_{n-j}^\circ}{a_n} \sim \left(\frac{w^2}{n}\right)^j. \quad (56)$$

There are  $a_j$  possible fragments of size  $j$ , each with the same probability. Conditional on that size, their distribution is exactly uniform. Equation (36) and the finite-shift expansion of (28a) supply their complete fixed-order probability expansions.

## 18 All orders and inverse growth for A007718

**Theorem 18.1.** *There are rational functions  $P_j^\circ(w)$  with  $P_0^\circ = 1$  such that for each fixed  $J \geq 0$  and some finite  $M_J^\circ$ ,*

$$c_n^\circ = e^{\Phi(n)} \left\{ \sum_{j=0}^J \frac{P_j^\circ(w)}{n^j} + O_J\left(\frac{(1+w)^{M_J^\circ}}{n^{J+1}}\right) \right\}. \quad (57)$$

*A finite rule computes them from the coefficients  $P_j$  in (28a) and the integers  $\rho_j$  in (35). In particular,*

$$P_1^\circ(w) = P_1(w) - w^2. \quad (58)$$

*Thus*

$$c_n^\circ \sim \frac{B_n^2}{n!} e^{w^2/2} \sim \frac{\exp\{n(w - \log w - 1 + 2/w) + w^2/2 - 2\}}{(w+1)\sqrt{2\pi n}}.$$

*Proof.* Insert (28a) at each of the finitely many shifts in (36). Taylor-expand the explicit smooth functions at  $n$ . Derivatives use  $W'(x) = W(x)/(x(1+W(x)))$ , so all resulting coefficients are rational functions of  $w$ . The leading shift factor is  $(w^2/n)^j$ , which means that only shifts  $j \leq J$  are relevant through order  $J$ . The omitted convolution term is

$$O_J(a_{n-J-1}) = e^{\Phi(n)} O_J(n^{-J-1}(1+w)^{2J+2}).$$

All other errors are the same power of  $n^{-1}$  times a fixed polynomial bound in  $w$ . No derivative of an asymptotic remainder is taken. The sole extra term at order one is  $\rho_1 w^2 = -w^2$ , proving (58).  $\square$

Explicitly, put  $F_J(x) = e^{\Phi(x)} \sum_{k=0}^J P_k(W(x))x^{-k}$  and expand  $\sum_{j=0}^J \rho_j F_J(x-j)/e^{\Phi(x)}$  through  $x^{-J}$ . This is the promised finite rule. For example the next coefficient obeys

$$P_2^\circ = P_2 - w^2 P_1 - \frac{w^3(3w^3 + 5w^2 + 3w + 2)}{(w+1)^2}.$$

For inversion define the explicit connected model

$$F_J^\circ(x) = e^{\Phi(x)} \sum_{j=0}^J P_j^\circ(W(x))x^{-j}. \quad (59)$$

It is positive and strictly increasing eventually, and  $(\log F_J^\circ)'(x) \sim \log x$ . Let  $x_J^\circ(y)$  be its tail inverse. The connected sequence itself is eventually increasing because (57) gives  $c_{n+1}^\circ/c_n^\circ \sim n/W(n)^2 \rightarrow \infty$ .

**Corollary 18.2.** *For the integer threshold  $N^\circ(y) = \min\{n \geq 1 : c_n^\circ \geq y\}$  there is a constant  $\tilde{C}_J > 0$  such that, for all sufficiently large  $y$ ,*

$$[x_J^\circ(y) - \varepsilon_J^\circ(y)] \leq N^\circ(y) \leq [x_J^\circ(y) + \varepsilon_J^\circ(y)], \quad \varepsilon_J^\circ(y) = \frac{\tilde{C}_J x_J^\circ(y)^{-J-1} (1 + W(x_J^\circ(y)))^{M_J^\circ}}{\log x_J^\circ(y)}. \quad (60)$$

Also  $x_J^\circ(c_n^\circ) = n + O_J(n^{-J-1}(1+w)^{M_J^\circ}/\log n)$ .

*Proof.* The relative error in (57) has the same size as the corresponding logarithmic error. Apply the mean-value theorem to  $\log F_J^\circ$ , whose derivative is asymptotic to  $\log x$ , and enlarge  $\tilde{C}_J$  to cover the two neighboring integer endpoints. Monotonicity then brackets the first index reaching  $y$ . The finitely many pre-tail values cannot affect the threshold for sufficiently large  $y$ .  $\square$

An unconditional single-ceiling formula would not follow near integer crossings. Equation (60) supplies the needed uncertainty interval; if its endpoint ceilings coincide, that common integer is the exact threshold. On the sequence range, nearest-integer rounding of  $x_J^\circ(c_n^\circ)$  eventually recovers  $n$ .

One may also invert compositionally. Let  $X$  solve  $\Phi(X) = \log y$ , and let  $h^\circ(X)$  be the formal logarithm of  $\sum_{j \geq 0} P_j^\circ(W(X))X^{-j}$ . The compatible formal inverse rule is

$$X + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left( \frac{1}{\Phi'(X)} \frac{d}{dX} \right)^{k-1} \frac{h^\circ(X)^k}{\Phi'(X)}.$$

At each requested algebraic order only finitely many terms are used; Taylor expansion applies to the finite model (59), and (60) controls the actual integer threshold.

[Added 5 October 2026, batch 100: Corollary 18.2 and the inverse rule are Part I's (30) and (31) with  $P_j$  replaced by  $P_j^\circ$ ; the corollary below is Part I's Proposition 8.1 applied to  $c_n^\circ$ , by the same argument. Both are printed as manuscript 02 gives them. ]

**Corollary 18.3.** *The positive-index sequence A007718 is not P-recursive, even over complex polynomial coefficients. Its ordinary generating function is not D-finite.*

*Proof.* The leading equivalent implies  $\log(n!/c_n^\circ) = 2n \log \log n + O(n)$ , so  $E^\circ(z) = \sum_{n \geq 1} (c_n^\circ/n!)z^n$  is entire. P-recursiveness of  $c_n^\circ$  would imply a polynomial-coefficient linear ODE for  $E^\circ$ . Every entire solution of such an ODE satisfies  $\max_{|z|=r} |E^\circ(z)| \leq \exp(Cr^M)$  for some constants  $C, M$ : outside a circle containing the zeros of the leading coefficient, write the companion first-order system and apply Gronwall along radial segments, since its rational matrix is bounded by a power of  $r$ . Cauchy's bound with  $r = (n/(CM))^{1/M}$  would force  $\log(n!/c_n^\circ) \geq (n/M) \log n - O(n)$ , a contradiction.  $\square$

## 19 Further questions and research

[Added 5 October 2026, batch 100: manuscript 02's section "Further research" is printed first, its three topics numbered as Questions II.1–II.3; Questions II.4 and II.5 record, under the standing rule for unproved claims, what Part II leaves existential or conditional. ]

- II.1. The positive residual gives a concrete route to the next symmetry coefficient: separate triple-leaf permutations, two leaf-pair swaps, higher-degree twin vertices and permutations of repeated components, then determine which classes contribute at order  $n^{-2}$  and account for their overlaps. The resulting group-sensitive expansion would refine the single-transposition law rather than only its remainder. [This is also the re-scoped Question I.2 of Part I.]

- II.2. The exact fragment formula (56) also invites a uniform analysis when the fragment size  $j = j(n)$  grows. This requires uniform shifted asymptotics and a quantitative transition between endpoint and central convolution contributions.
- II.3. A further practical goal is to make the Bell-ratio, weighted-tail and smooth-convexity constants explicit, producing certified finite- $n$  probability bounds and usable onset estimates for the inverse brackets.
- II.4. *Explicit exponents* [write]. The exponents  $\xi_1$  in (37),  $\xi_2$  in (55),  $\xi_3$  in the estimate of  $a_{n-2}/a_n$ , and  $M_j^\circ$  in Theorem 18.1 and Corollary 18.2 are shown to exist (as “some fixed finite” exponent, or as “a fixed polynomial in  $w$ ”) and are not computed. They depend on Part I’s  $M_j'$ , itself not computed (Question I.6). Missing: explicit values, at least for  $\xi_1$ , which governs the third term of the disconnection probability.
- II.5. *Dependence on Part I* [write]. Manuscript 02 states that it does not re-audit the analytic estimates of Report 88 (its `02-symmetry-verification.md`). Every theorem of Part II therefore rests on Part I’s Theorems 1.1 and 1.2, the expansion (28a) at orders  $J \leq 2$ , and the shifts (17), including the steps of Part I recorded as Questions I.5–I.7. Part II’s own arguments use these only at fixed low order; a written proof of Part I’s sketched steps at those orders would make Part II self-contained.

## 20 Exact checks and source scope

The direct checker constructs all pairs of set partitions through  $n = 6$ , converts them to integer matrices, and canonicalizes by row and column permutations without merging the shores. It enumerates 44,169 partition pairs and 438 graph isomorphism classes in total, independently computing vertex automorphism orders, leaf transpositions, components and isolated edges. A second checker examines 226,342 edge permutations to verify every parallel-edge-kernel fiber, the leaf-lift counts, the unique-giant identity and every fragment-size count. A cycle-type checker verifies the marked mass, reciprocal coefficients and Euler transform on 2,714 cycle types through  $n = 20$ , agreeing with the independent graph census at small sizes. Together the checkers test (4), (43), the residual gap, the isolated-edge bijection, and the connected values. These are exact finite-identity and transcription checks, not proofs of asymptotic inequalities or effective onset.

[Added 5 October 2026, batch 100: the three checkers are shipped as `code/02-symmetry-check_graphs.py`, `code/02-symmetry-check_kernel_and_giant.py` and `code/02-symmetry-check_cycles.py`, their recorded outputs in `data/`; the intake reran them on copies on 5 October 2026, and every output equals the recorded one apart from line endings. The first reciprocal coefficients recorded there are 1, −1, −3, −3, −8, −8, −38. ]

The component method is classical in scope: Wright [7] and Bell, Bender, Cameron and Richmond [8] analyze connectedness through counting sequences and component generating functions. Here we prove the needed endpoint and no-giant estimates directly for the present growth scale and retain the finite-shift remainder at every fixed order. The 2025 study by Cameron, del Valle and Roney-Dougal [9] considers regular bipartite multigraphs with fixed shore cardinality and increasing common degree. Its parameter regime and automorphism convention differ from this uniform variable-vertex-count ensemble. It is not a substitute for the kernel cancellation above.

This is an unrefereed mathematical companion to Part I, with reproducible exact checks, not a Lean formalization. The preceding all-orders estimates are included as an explicit dependency. The literature search is bounded, and no exhaustive or first-in-literature priority assertion is made. No claim of effective constants, numerical asymptotic onset, or classification of all exponentially small corrections is intended.

[Added 5 October 2026, batch 100: “included as an explicit dependency” refers to the delivered package, which shipped Report 88’s archive unchanged; here that archive is not shipped again, since its content is Part I and its files (see Appendix A). ]

## A Provenance and editorial record

Written 5 October 2026 (batch 100).

**Sources.** Two manuscripts, both from the session bundle of Research Reports 1–243 that arrived in ProveIt commit 60f54ea06 (5 October 2026), placed by 36571ae0e as batch-100 manuscripts 01 and 02. Neither names a ProveIt commit, path or pin; neither continues a repository report.

- Manuscript 01, Report 88, `ProveIt_A007716_Asymptotics_and_Inverses.zip` (17 files in `A007716_Asymptotics/`; `article.tex`, 469 lines, 37 labels; 10-page PDF *Asymptotic Expansions for Unlabeled Bipartite Multigraphs: OEIS A007716 and its inverse growth*, dated 1 October 2026). Printed as Part I, Sections 1–11.
- Manuscript 02, Report 92, `ProveIt_A007716_Symmetry_and_A007718_Connectivity.zip` (14 files in `A007716_Symmetry_and_A007718_Connectivity/`; `article.tex`, 414 lines, 44 labels; 10-page PDF *Rare Symmetries and Connected Bipartite Multigraphs: Probability laws for A007716 and asymptotic expansions for A007718*, dated 2 October 2026). Printed as Part II, Sections 12–20. Its `dependency/` directory held Report 88’s archive, byte-identical to manuscript 01’s.

Both author lines and PDF metadata read “Research note prepared for Vladimir Reshetnikov with OpenAI”; neither says “prepared for private review”.

**Where the merge chose.** Base: Report 88, because Report 92 depends on it and calls it the preceding report; the two prove different theorems, so neither is more general. Displays that both print are printed once, in Part I: manuscript 02’s (Z) and (P1) are Part I’s (4) and (29); its (allinput), (burn), (ewens) and (bellshift) are (28a), (6), (20) and (17). Its (inputs), (markbound) and (goodT) are kept as Part II’s (38), (39) and (40), with notes, because Part I does not display them in that form. Its (phider) is kept inside its proof (it adds  $\Phi'' \sim 1/x$ ). Its corollary on non-P-recursiveness and its compositional inverse are kept although they repeat Part I’s arguments for  $c_n^o$  (note before Corollary 18.3). The two bibliographies are merged: manuscript 01’s OEIS and manuscript 02’s OEIS716 are one entry; manuscript 02’s *Prior* (Report 88) is replaced by references to Part I.

**Labels and numbers.** Part I’s 37 delivered labels carry the prefix `bpm:`, unchanged otherwise, and its section, theorem and equation numbers are those of manuscript 01. Of manuscript 02’s 44 labels, 38 carry the prefix `bpm:sym::`; the six of the duplicated displays `eq:Z`, `eq:allinput`, `eq:P1`, `eq:burn`, `eq:ewens` and `eq:bellshift` are not reprinted. Part II’s section and theorem numbers are manuscript 02’s plus eleven; its equation numbers continue Part I’s. The write added the labels of the two Parts, the conventions section and table, this appendix, Part I’s display (28a) (unnumbered in manuscript 01; tagged (28a) so that every other equation keeps manuscript 01’s number), its Propositions 8.1 and 9.1, its Sections 6, 7, 9 and 10, Part II’s Sections 12, 19 and 20 and Corollary 18.3, and the question items: 19 labels in Part I and 10 in Part II, 104 in all.

**Renamed symbols.** All in Part II; Table 1 lists them:  $c_n \rightarrow c_n^\circ$ ,  $b_j \rightarrow \rho_j$ ,  $A(z) \rightarrow \mathcal{A}(z)$ ,  $A_1 \rightarrow \alpha_1$ ,  $D(w) \rightarrow \delta(w)$ ,  $R_n \rightarrow \mathcal{R}_n$ ,  $L(G) \rightarrow \Lambda(G)$ ,  $K_G \rightarrow \Omega_G$ ,  $K \rightarrow q$ ,  $K_J \rightarrow \tilde{C}_J$ ,  $C_j \rightarrow P_j^\circ$ ,  $N_J \rightarrow M_J^\circ$ ,  $G_J \rightarrow F_J^\circ$ ,  $x_J \rightarrow x_J^\circ$ ,  $\varepsilon_J \rightarrow \varepsilon_J^\circ$ ,  $N_c \rightarrow N^\circ$ ,  $h \rightarrow h^\circ$ ,  $E(z) \rightarrow E^\circ(z)$ ,  $F_n(t) \rightarrow \Psi_n(t)$ ,  $N \rightarrow n_0$ ,  $H \rightarrow \Gamma$  (a fragment class),  $v(H) \rightarrow |V(\Gamma)|$ ,  $H(p) \rightarrow \text{Ent}(p)$ ,  $r(G) \rightarrow \varrho(G)$ ,  $m = |\text{Aut}(G)| \rightarrow g$ ,  $M, M', M'' \rightarrow \xi_1, \xi_2, \xi_3$ , the restated  $M_J \rightarrow M'_J$ ,  $r_i \rightarrow a_i$ , and the restated  $\Phi$ 's  $u \rightarrow w$ . No normalization changed.

**Text added by the write.** The editorial note, the conventions section, the Part headings and this appendix; the notes marked “[Added 5 October 2026, batch 100]”: in Part I after Theorem 1.2, after (5) (the labelled analogue A120733), after the Bell display (16) (the repository’s earlier proof of its order-one case), after (25), after (26) (the independent check of the unmarked part of  $Q_2$ ), at Question I.2 (answered at leading order and re-scoped), before Questions I.5–I.8, and in Section 11; in Part II at its head, in Section 13, after the proof of Theorem 12.1, after Lemma 16.1, before Corollary 18.3, in Sections 19 and 20. The bracketed sentence at the end of Question II.1 is also the write’s. The abstracts are the two manuscripts’ abstracts. Nothing of either manuscript is omitted except the six duplicated displays named above, manuscript 02’s title block and its bibliography entry for Report 88.

**Standing rule.** No claim of either manuscript was found to be wrong. The intake’s checks (README) confirm every printed coefficient it tested:  $Q_1 = R_1 + w^3$ ,  $P_1 = Q_1 + 2\beta_1 - 1/12$ ,  $\beta_1$  from the Gaussian correction, the identity (28),  $\delta(w)$ ,  $w^2\delta + 3w^4$ ,  $P_2^\circ$ , and numerical agreement of  $Z_n$ ,  $Y_{n-2}$  and  $Z_n + Y_{n-2}$  with the predicted corrections. Steps that are sketched, asserted without derivation, or existential are Questions I.5–I.8 of Part I and II.4–II.5 of Part II.

**Neighbouring reports.** In the collection `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`: `a260700-parabolic-double-cosets` names the labelled analogue A120733 of  $\sum M_{k,l}(n)$  (note after (5)); `a277364-bell-asymptotics` proves the order-one Bell expansion (15) (note in Section 4); `a307316-leafless-multigraphs`, a batch-100 report, treats unlabelled loopless multigraphs with minimum degree two on the scale  $W(2m)$ , by a different method (Wright’s simple-graph theorem and a component convolution), and leaves a sharp unlabelled equivalent open (its `lfm:q:sharp`, which names the exact invariant-partition formula of Part I and the automorphism estimate of Part II as a possible route); its `lfm:prop:deficit` uses the leafless counterpart of Part II’s isolated-edge bijection, conditionally on a relative equivalent, and its upper bound `lfm:eq:supportmain` carries the analogous factor  $e^{w^2/2}$  on the scale  $w = W(2m)$ ; `a088714-bell-scale-growth` compares another sequence, A088714, with the Bell numbers. No theorem is shared with any of them.

**Independent check of the batch-100 write (7 October 2026).** An adversarial check made by the intake after the write (commit 72091374c) re-derived every statement the write added, with its own code, from fresh extractions of both archives. *Sources.* Archive sizes, file counts, line and label counts, page counts, titles, dates and author lines are as printed; Report 92’s `dependency/` archive is byte-identical to Report 88’s; Part II begins at Section 12 and its Theorem 1.1 is Theorem 12.1; the label counts (37 and 19 in Part I, 38 of 44 and 10 in Part II, 104 in all, the six omitted ones being the duplicated displays) and the decimal numerals of both texts (unchanged from the manuscripts) were recounted. OEIS A007716 (revision #55), A007718 (#18) and A120733 (#102) were fetched again: A007716 has a cycle-index formula and no asymptotic; the Euler transform of A007718 with  $c_0^\circ = 0$  gives A007716 for  $n \leq 50$ ; the reciprocal coefficients of  $\mathcal{A}(z)$  are 1,  $-1$ ,  $-3$ ,  $-3$ ,  $-8$ ,  $-8$ ,  $-38$ ; and  $\sum_{k,l} M_{k,l}(n)$ , computed by inclusion–exclusion, is A120733 for  $n \leq 8$ . The cited passages of `a277364-bell-asymptotics`, `a260700-parabolic`

`-double-cosets` and `a307316-leafless-multigraphs` say what the notes say (the Gaussian correction plus  $\frac{1}{12}$  is  $\beta_1$  by direct expansion), and Pavlichin–Jiao–Weissman’s Theorem 12, proved in their Appendix C, states  $e^{(1+o(1))n \log n}$  after the words “precise asymptotics”. *Statements.* Every identification in the notes was rechecked by hand:  $\tau(i) - 1 = E(b)$  for one selected cycle, the two cutoff orders  $\log^{3J+3} n$  and  $w^{4J+4}$  of Question I.6,  $Q_2 \sim w^8/288$ ,  $n/(2w) = x/2$ ,  $w = \log n - \log \log n + o(1)$ , the second line of (38), the identity  $\mathcal{R}_n = \sum_G \varrho(G)$  and the agreement of (50) with (28). *The check of  $Q_2$ .* In `data/01-asymptotics-reconstruct_Q2.json` the four contributions without marks add up to the printed  $R_2$  (SymPy), and the seven to  $Q_2$ . By a route independent of the mass formula,  $Z_n/(B_n^2/n!) = \sum_d \left[ \begin{smallmatrix} n \\ n-d \end{smallmatrix} \right] (B_{n-d}/B_n)^2$  with exact Stirling numbers of the first kind and Bell numbers to 40 digits, the ratio of  $n^2 \{ Z_n/((B_n^2/n!)e^{w^2/2}) - 1 - R_1(w)/n \}$  to  $R_2(w)$  is 1.04124, 1.04000, 1.02454, 1.01443, 1.00760, 1.00404 at the six sizes of the note after (26), confirming its numbers. Against the exact  $a_n$  of the OEIS b-file ( $n \leq 50$ , where  $w \leq 2.9$ ), the full  $Q_2$  cannot be tested: even  $n(a_n/((B_n^2/n!)e^{w^2/2}) - 1)/Q_1$  is still 1.75 at  $n = 50$ . No mathematical error and no defect of the write was found.

## References

- [1] OEIS Foundation Inc., A007716, consulted 1 October 2026 (Part I) and 2 October 2026 (Part II). <https://oeis.org/A007716>.
- [2] OEIS Foundation Inc., A007718, consulted 2 October 2026. <https://oeis.org/A007718>.
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- [4] P. J. Cameron, T. Prellberg and D. Stark, *Asymptotics for incidence matrix classes*, Electron. J. Combin. **13**(1) (2006), R85. <https://arxiv.org/abs/math/0510155>.
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